BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

SECTION- A Multiple Choice Questions

Note: Q.1 Choose the correct answer for each form the given options:

Q1. 3(5+7) = 3x5+ 3x7, name the property used.

(a) commutative
(b) Associative
(c) Distributive
(d) Closure

Q2. The characteristic of log 0.000548 is _____.

(a) 4
(b) -4
(c) 3
(d) -3

Q3. The degree of the 3x2y+5y4-10 is _____.

(a) 4
(b) 5
(c) 6
(d) 10

Q4. The diagonal of _____ does not divide it into two congruent triangles:

(a) Rectangle
(b) Trapezium
(c) Parallelogram
(d) Square

Q5. a2 + b2 + C2 + 2ab - 2ca is equal to:

(a) (a+b+c)2
(b) (a+b-c)2
(c) (a-b+c)2
(d) (a-b-c)2

Q6. Factors of a2 +2a - 24 are:

(a) a+4, a-6
(b) a-4, a-6
(c) a-8, a+3
(d) a+8, a-3

Q7. HCF of x3 - y3 and x2 + xy - y2 is:

(a) x+y
(b) x-y
(c) x2-xy+y2
(d) x2+xy+y2

Q8. The_____ of a triangle cut each other in the ratio 2:1.

(a) Altitude
(b) Median
(c) angle bisector
(d) right bisector

Q9. Tthe solution set of sqrt(5x+5)+5 = 10 is:

(a) [illegible in source scan]
(b) {5}
(c) {4}
(d) {-4}

Q11. The points (-3,4) lies in_____ quadrant.

(a) First
(b) second
(c) Thirds
(d) fourth

Q12. Perpendicular form vertex to a triangle to its opposite side is called_____

(a) Median
(b) Perpendicular besector
(c) Altitude
(d) angle

SECTION - B

> Note: solve any EIGHT of the following questions, each question carries 04 marks.

Q1. simplify (2+6i)/(3-i) (4+i)/(3+i)

Q2. Solve: [illegible in source scan]

Q3. find the continued product by using relevant formula: (X-Y) (X+Y) (X2 +Y2)-(X4 +Y)

Q4. simplify [illegible in source scan]

Q5. find the continued product by sung relevant formula: [illegible in source scan]

Q6. Find remainder by using remainder theorem X3-3x+4X-14 is deivded by (x+2)

Q7. find square root or 25x4+40x3 + 26x2 + 8x +1

Q8. Prove that. Any point on the bisector of an angle is equidistant from its arms.

Q9. Find the solution set of (2x+1)/3 =8

Q10. solve by using quadratic formula 10x2 +19x - 15 =0

Q11. If x=3-2sqrt2 then find x2 + 1/x2

Q12. solves: (7X-4)/15 = (7 X+4)/10

SECTION - C

> Note: Solve any FOUR of the following questions, Each question carries 07 marks.

Q13. Factorize any three of the following questions. Each question carries 07 marks. (i) 4a2 + 12 ab +9ab2 (ii) y2+7y - 98 (iii) x4 + 4y4 (iv) (x2 = 4x - 5)(x2 - rx-12) - 144

Q14. Solve the equation x +sqrt(x) + 5 = 7

Q15. Solve path the hell of logarithm (4.398)3 x sqrt(80.06) / log 3342/5

Q16. Find the value of a2 - b2 and ab when a-b==5 and a-b-3.

Q17. Show that the punts A (3, 2), B(9.10) and c(1.16) form an Isoscels triangle.

Q18. Construct a triangle ABC such that MAB=5 cm, m BC - 5-5 cm and m>A=45 degrees